Graph-Equation Stability and Certified Shortening in Positive Scalar Curvature

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Abstract

We study the graph equation and the finite shortening construction in *Positive scalar curvature forces rational inessentiality*. The global rational-inessentiality assertion is an upstream claim, not a consequence of a finite inverse or of the calculations in this report. On the source's uniformly controlled covers we replace monotonicity of the graph profile by strict radial flux ellipticity, including explicit nonmonotone examples. A tangent-chord comparison gives a nonlinear arctangent residual bound for heights. For curvature, we retain the directional derivative of the equation residual and complete the two source squares, obtaining an explicit quadratic residual penalty that can be summed through a finite shortening pass in the original fibre rank. A smooth periodic counterexample shows why small equation residuals and even convergence in two derivatives cannot replace that additional control. The Ram weak-operator and nonlinear-stability proofs are used with their actual domains and boundary conditions, not as a substitute for cover estimates, degree realization, or the torical obstruction. Contribution assessment: C3/5, source-specific flux assumptions and nonlinear height and curvature error bounds give a notable local extension while leaving the global geometric obligations explicit.

1 Problem and Scope

For a closed connected oriented smooth n-manifold M, rational inessentiality means

$$(c_M)_*[M] = 0 \quad \text{in } H_n(B\pi_1(M); \mathbb{Q}),$$

where c_M classifies the universal cover. The assigned source [1] claims that positive scalar curvature implies this vanishing for every $n \geq 2$, without a spin or group hypothesis. It also derives aspherical obstructions, nonnegative-curvature rigidity, and simplicial-volume vanishing. This report does not independently establish those global conclusions.

The useful analytic interface is unusually concrete. A scalar coordinate ℓ on a cover is replaced by εf. Increasing the metric in the df direction contracts that new coordinate, and a compensating warped circle preserves volume and produces favorable scalar terms. The equation selecting f, its bounded solution class, and the scalar loss must all be retained. Neither an invertible finite response matrix nor a weak residual in a few tests supplies those three ingredients.

Our new results are Theorems 3.1, 4.2, 5.1, and 6.1. The first is an assumption extension of the source's analytic proof; the next three are quantitative deductions with complete local proofs. Maximum principles, tangent convexity, and completion of squares are classical tools. No historical priority or optimal global scalar-curvature constant is asserted.

2 The Actual Graph Operator

Let B be a Riemannian cover of a closed manifold, possibly times a fixed finite flat torus. Use lifted nested charts from a finite atlas. A uniformly controlled metric g is uniformly equivalent to the complete reference metric, with all coefficient derivatives uniformly bounded. The norms $C_b^{j,\alpha}$ are suprema of chart norms, with enlarged charts available for interior estimates; $0 < \alpha < 1$. Every assertion below concerns a fixed finite-dimensional geometry. There is no uniformity in the number of added circles unless explicitly stated.

Fix $\varepsilon, w, B_0, s > 0$, $0 \le \beta < 1$, and a positive smooth profile $N : [0,\infty) \to (0,\infty)$, constant near zero and equal to one for $t \ge 1/2$. For a real function f put

$$\begin{gathered} t = |df|_g\,, \quad v = (1 + N(t)^2 t^2)^{-1/2}, \quad U = vN(t)\nabla f, \quad r = |U|_g\,, \quad D = -\operatorname{div}_g U, \\ \lambda(r) = B_0(r^2 + s^2)^{\beta/2}, \quad \theta_f = (\ell - \varepsilon f)/w, \quad F_\ell(f) = D - \lambda(r)\tan\theta_f. \end{gathered} \tag{1}$$

The admissible class has $f \in C_b^{2,\alpha}$ and a positive uniform margin from both endpoints of $(-\pi/2, \pi/2)$ for θ_f. The residual $F_\ell(f)$ is a scalar function, not an unscaled vector of moment equations. Although t need not be smooth at $df = 0$, the flux and $\lambda(r)$ are smooth functions of df, by the constant-profile condition near zero.

The linearization, with every coefficient evaluated at f, is

$$DF_\ell(f)\xi = -\operatorname{div}_g(a_f\, d\xi) - \lambda'(r)\tan\theta_f\, Dr_f[d\xi] + \frac{\varepsilon\lambda(r)}{w}\sec^2\theta_f\,\xi, \tag{2}$$

where a_f is the derivative of the covector-to-vector flux. The middle expression is interpreted smoothly through $r = 0$ by differentiating λ as a function of r^2. It is a first-order drift, not a term that can be dropped in Newton's method. Its presence also means that a symmetric coercive matrix surrogate is not the full continuous linearization.

2.1 Exhaustion with the Correct Solution Class

On a bounded-gradient, positive-margin set the principal eigenvalues are

$$a_\perp(t) = \frac{N(t)}{\sqrt{1 + N(t)^2 t^2}}, \qquad a_\parallel(t) = \frac{N(t)(1 + \mathfrak{p}(t))}{(1 + N(t)^2 t^2)^{3/2}}, \qquad \mathfrak{p}(t) = tN'(t)/N(t). \tag{3}$$

The zeroth-order coefficient in (2) is at least $c_0 = \varepsilon B_0 s^\beta / w > 0$. The source's linear inverse uses zero Dirichlet data on smooth compact exhaustions, the bound $\|\xi\|_\infty \le c_0^{-1}\|L\xi\|_\infty$, and interior Schauder estimates on every fixed enlarged chart. It does not require uniform boundary Schauder constants on the moving exhaustion. Diagonal compactness then supplies a bounded global solution.

Uniqueness is in $C_b^{2,\alpha}$. For a positive supremum of a homogeneous solution, recenter uniform charts at almost-maximizers and pass to a $C^{2,\alpha'}$ local limit, $0 < \alpha' < \alpha$. At the limiting maximum the drift vanishes, the Hessian has the correct sign, and $L\xi \ge c_0 \sup\xi > 0$. This proves the needed maximum principle without assuming the supremum is attained on the cover. The compact-domain Dirichlet and interior regularity theorems remain classical analytic inputs [2].

There is no arbitrary growth class at infinity here. On $\mathbb{R}$, $-u'' + u = 0$ has the unbounded solution e^x, while on $(-R, R)$ the solution with boundary values e^R is $e^R \cosh x / \cosh R$, whose value at zero tends to 2. Thus increasing domains alone does not erase boundary data that grow. Zero boundary data and a bounded global class are material conditions, not a choice of notation.

3 Allowing Nonmonotone Profiles

Theorem 3.1 (Radial ellipticity replaces profile monotonicity)**.** *Retain the geometry, positive parameters, smooth bounded ℓ, and the constant-near-zero and $N = 1$ for $t \geq 1/2$ conditions above. Replace the source requirement $N' \geq 0$ by*

$$1 + \mathfrak{p}(t) > 0 \qquad (t > 0). \tag{4}$$

Then the source's uniform graph-solvability argument still gives a unique admissible solution of $F_\ell(f) = 0$, smooth with bounded derivatives of every order. Its symmetry and smooth-parameter conclusions remain valid under the source's common mixed-derivative bounds. For scalar nonnegativity of the additional transverse term, it suffices in addition that

$$\mathfrak{p}(t)N(t)^2t^2 \leq 1. \tag{5}$$

This extension of the graph equation alone does not establish the source's logarithmic shortening estimate for arbitrary such profiles.

Proof. The two eigenvalues in (3) are positive. On every fixed compact gradient interval their minimum is positive, including zero by smooth extension. The linear inverse and openness in the homotopy $\ell \mapsto \tau\ell$, $0 \leq \tau \leq 1$, are unchanged. The strict zeroth-order coefficient and comparison do not use $N' \geq 0$.

The barriers

$$f_\pm = \varepsilon^{-1}\{\tau\ell \pm (\pi/2 - \delta)w\}$$

have uniformly bounded gradient and divergence along the homotopy. Their tangent terms tend with opposite signs to infinity as $\delta \downarrow 0$, while $\lambda \geq B_0 s^\beta > 0$. They supply a fixed uniform band margin and height bound. All values and derivatives of N needed on their gradient ranges are bounded by compactness, without monotonicity.

For the gradient estimate, first bootstrap each individual $C_b^{2,\alpha}$ solution using its own ellipticity and band margin. Recenter at almost-maximal slopes and take a smooth local limit. If the maximum slope exceeds one, $N = 1$ near that point. The source's ordinary-graph Jacobi identity is therefore unchanged. With

$$h_e = \lambda \sec^2 \theta_f / w, \qquad 0 < h_{\min} \leq h_e \leq h_{\max},$$

it gives

$$\varepsilon h_{\min}/2 \leq v\{C_{\mathrm{Ric}} + h_{\max} \|d\ell\|_\infty\}.$$

If the brace is zero the asserted large-slope maximum is impossible; otherwise this gives a positive lower bound for v. The bounded-slope case is already controlled. Thus there is a gradient bound independent of τ.

After that bound, (4) supplies one ellipticity constant. Differentiate the divergence equation, apply interior scalar Hölder regularity to the first derivatives, and then interior Schauder estimates and bootstrapping. These are the same classical estimates used in the source; their data are bounded on the attained gradient and band ranges. Local compactness on an exhaustion gives closedness in τ. Openness, closedness, and $f = 0$ at $\tau = 0$ finish existence. The limiting-chart comparison proves uniqueness. Naturality then proves isometry invariance; the Banach implicit-function theorem with common mixed bounds gives parameter smoothness and compatibility with fixed deck transformations. None of these steps uses the sign of N' separately.

For the last assertion the transverse coefficient in the source scalar identity is $2r^2(1+\mathfrak{p})(v^2-\mathfrak{p}r^2)$. The first factor is positive by (4), and

$$v^2 - \mathfrak{p}r^2 = v^2(1 - \mathfrak{p}N^2t^2) \geq 0.$$

The remaining claims follow. Constants may deteriorate as $\inf(1+\mathfrak{p})$ on a relevant interval tends to zero. □

3.1 A Genuine Nonmonotone Example

Take a nonzero nonnegative $\psi \in C_c^\infty((1/8, 1/4))$ and set

$$K = \max\{1, \|\psi\|_\infty, \|t\psi'\|_\infty\}, \qquad N(t) = \exp(\psi(t)/(8K)).$$

Then $1 \leq N \leq e^{1/8}$, $|\mathfrak{p}| \leq 1/8$, and N rises and falls, so it is not nondecreasing. On its support

$$\mathfrak{p}N^2t^2 \leq e^{1/4}/128 < 1;$$

elsewhere $\mathfrak{p} = 0$. All assumptions of Theorem 3.1 hold. This example also separates ellipticity from arbitrary oscillation: a positive profile with $tN'(t)/N(t) < -1$ at some t has a negative radial principal eigenvalue there.

4 A Nonlinear Height Certificate

Lemma 4.1 (Tangent chord)**.** *If $a, b \in (-\pi/2, \pi/2)$ and $a - b = d \geq 0$, then*

$$\tan a - \tan b \geq 2\tan(d/2). \tag{6}$$

For a fixed $d < \pi$, equality holds at $a = d/2,\ b = -d/2$.

Proof. Writing $m = (a+b)/2$,

$$\tan a - \tan b = \frac{\sin d}{\cos a \cos b}, \qquad \cos a \cos b = \cos^2 m - \sin^2(d/2) \leq \cos^2(d/2).$$

For $0 < d < \pi$ the numerator and denominator are positive, giving (6); the zero case follows directly. □

Theorem 4.2 (Residual and data comparison)**.** *Use a common metric, flux profile satisfying (4), and positive parameters. Let $f_1, f_2 \in C_b^{2,\alpha}$ have uniform tangent-band margins for data ℓ_1, ℓ_2. Put*

$$e_i = F_{\ell_i}(f_i), \quad \rho_i = \|e_i/\lambda(r_i)\|_\infty, \quad \Delta = \|\ell_1 - \ell_2\|_\infty.$$

Then

$$\|f_1 - f_2\|_\infty \leq \frac{\Delta}{\varepsilon} + \frac{2w}{\varepsilon}\arctan\frac{\rho_1 + \rho_2}{2}. \tag{7}$$

In particular, for an exact solution f and an admissible approximate solution $\widehat{f}$ to the same data, if $\left\|F_\ell(\widehat{f})\right\|_\infty \leq \eta$, then

$$\left\|\widehat{f} - f\right\|_\infty \leq \frac{2w}{\varepsilon}\arctan\frac{\eta}{2B_0 s^\beta} \leq \frac{w\eta}{\varepsilon B_0 s^\beta}. \tag{8}$$

Existence of the exact solution is not inferred from this estimate.

Proof. If $d = \sup(f_1 - f_2) > \Delta/\varepsilon$, use limiting uniform charts at almost-maximizers. Bounded $C^{2,\alpha}$ norms and smooth fixed coefficients permit a common local limit. At its center the gradients agree, the Hessian of $f_1 - f_2$ is nonpositive, and flux ellipticity implies $D_1 - D_2 \geq 0$. Also $r_1 = r_2$ and the common λ is positive. The angle difference satisfies

$$\theta_2 - \theta_1 = (\ell_2 - \ell_1 + \varepsilon d)/w \geq (\varepsilon d - \Delta)/w > 0.$$

Subtracting the two residual equations and dividing by the common λ gives

$$\rho_1 + \rho_2 \geq (e_1 - e_2)/\lambda \geq \tan\theta_2 - \tan\theta_1 \geq 2\tan((\varepsilon d - \Delta)/(2w)).$$

Invert the increasing tangent. If $d \leq \Delta/\varepsilon$, the asserted bound is automatic. Interchange the two functions to bound the other sign. The uniform band margins justify all limiting trigonometric expressions. Finally $\lambda \geq B_0 s^\beta$ and $\arctan x \leq x$ give (8). □

On a compact smooth domain the same proof gives the maximum of the right side of (7) and $\|f_1 - f_2\|_{L^\infty(\partial\Omega)}$. Boundary discrepancies do not disappear from a truncated solve. Exact solutions also satisfy the data stability bound $\|f_1 - f_2\|_\infty \leq \Delta/\varepsilon$. For $\varepsilon = w = B_0 = s = 1$, $\beta = 0$, a constant approximate field c for $\ell = 0$ has residual $\tan c$, and the exact field is zero. Its actual error is $|c| = \arctan|F_0(c)|$. This example checks signs but does not attain the symmetric chord bound. Sharpness in Lemma 4.1 is scalar sharpness, not a claim of optimality over all unforced geometric solutions.

5 Curvature Needs a Different Error Budget

The graph and compensating circle give

$$g^+ = g + N(t)^2 df^2 + v^2 d\vartheta^2, \qquad (g + N^2 df^2)^{-1} = g^{-1} - U \otimes U, \quad dV_{g^+} = dV_g\, d\vartheta. \tag{9}$$

The last equality is exact: the graph determinant contributes v^{-2} and the circle contributes v^2. All differential norms below use the current physical metric, not coefficient Euclidean norms.

For $t > 0$, choose $e_1 = \nabla f/t$ and complete it to an orthonormal frame. Set $z = e_1 r$, $A = d\log t$, and $C_{jk} = t^{-1}\operatorname{Hess} f(e_j, e_k)$ for $j, k > 1$. The source computes the exact identity

$$\begin{aligned}\operatorname{Scal}_{g^+} - \operatorname{Scal}_g = {} & D^2 - 2U(D) + r^2|C|^2 + z^2 \\ & + 2r^2(1+\mathfrak{p})(v^2 - \mathfrak{p} r^2)\sum_{j>1} A(e_j)^2.\end{aligned} \tag{10}$$

Its convention is $\Delta = \operatorname{div}\nabla$, positive scalar curvature on round spheres, and $\mathrm{II}(X, Y) = \langle \nabla_X \nu, Y\rangle$. The sign in $-2U(D)$ is particularly important: absolute smallness of $D - \lambda\tan\theta$ says nothing about its derivative along U. We use (10) as a source geometric identity and check its subsequent algebra; no finite test is a proof of the Gauss and warped product formulas.

Theorem 5.1 (A residual-sensitive scalar bound)**.** *Let f be smooth, with the quantities in (1), and suppose $(1+\mathfrak{p})(1 - \mathfrak{p}N^2t^2) \geq 0$. Define the actual residual and the adverse height derivative by*

$$e = D - \lambda\tan\theta_f, \qquad A_+ = (U\ell - \varepsilon r t)_+.$$

At a point where $t > 0$, set

$$\gamma = \frac{\beta r^2}{r^2 + s^2}, \qquad b = \frac{2A_+}{w\lambda}, \qquad c = \gamma^2 + b. \tag{11}$$

If $c < 1$, then

$$\mathrm{Scal}_{g^+} \geq \mathrm{Scal}_g - \frac{2\lambda A_+}{w} - \frac{c}{1-c}e^2 - 2(Ue)_+. \tag{12}$$

The coefficient $c/(1-c)$ is the exact optimal coefficient obtained by eliminating the two scalar variables D, z in this quadratic estimate. This is not a geometric optimality assertion.

Proof. Since $U(f) = rt$ and $U(\lambda)/\lambda = \gamma z$, differentiating $D = \lambda \tan\theta_f + e$ gives exactly

$$U(D) = \gamma z(D-e) + \frac{1}{w}\left(\lambda + \frac{(D-e)^2}{\lambda}\right)(U\ell - \varepsilon rt) + U(e). \tag{13}$$

The bracket is nonnegative. Discard the nonnegative C and transverse terms in (10), and replace $U\ell - \varepsilon rt$ by A_+. Apart from $-2\lambda A_+/w - 2Ue$, the expression is

$$Q = D^2 + z^2 - 2\gamma z(D-e) - b(D-e)^2.$$

There is an exact identity

$$Q = \{z - \gamma(D-e)\}^2 + (1-c)\left(D + \frac{ce}{1-c}\right)^2 - \frac{c}{1-c}e^2. \tag{14}$$

This proves the bound. Choosing $D = -ce/(1-c)$ and $z = \gamma(D-e)$ makes both squares zero, which proves the stated algebraic optimality. Finally $-2Ue \geq -2(Ue)_+$. □

The estimate extends by continuity across a zero of df whenever a uniform $c \leq c_* < 1$ is available. At an interior point of the set $df = 0$, the metric simply acquires a constant circle; $U = 0$ and the nonnegative error allowance is harmless. Thus no division by $t = 0$ is made in the final global bound.

5.1 An Exact Rational Budget

For $\gamma = 1/2,\ b = 1/4$ we have $c = 1/2$. If $\lambda A_+/w = 1/100$, $|e| \leq 1/10$, and $(Ue)_+ \leq 1/200$, (12) loses at most

$$2/100 + 1/100 + 2/200 = 1/25.$$

For $e = 1/10$, the quadratic worst case is $D = -1/10,\ z = -1/10$, where $Q = -1/100$. These are compatible algebraic budgets at the stated interface, not a constructed complete positive-curvature manifold. Using an actual pointwise c can be substantially better than replacing it by a uniform bound close to one.

6 A Finite Inexact Shortening Pass

We now specialize to the source profile and parameters so that the global coordinate-count dependence remains the source's, rather than a new unproved claim about arbitrary nonmonotone profiles. Let

$$\begin{gathered} q \geq 2, \quad a = (2 + \log q)^{-1}, \quad \beta = 1 - a, \quad s = q^{-3}, \\ \varepsilon = L/4, \qquad B_0 = 4L/(aw), \\ N(t)^2 = s^2 + (1-s^2)\phi(t), \end{gathered} \tag{15}$$

where $0 \leq \phi' \leq 10$, $\phi = 0$ on $t \leq 1/4$, and $\phi = 1$ on $t \geq 1/2$. Then $\mathfrak{p}N^2t^2 = \frac{1}{2}t^3(1-s^2)\phi' \leq 5/8$. Let g_* be the original n-dimensional fibre metric at the start of a pass. Added circles are separate warped blocks. All coordinates and graph functions are assumed exactly torus invariant.

Theorem 6.1 (Shortening with an explicit total residual cost)**.** *Assume the source's finite-stage bounded geometry and, for each original coordinate, $|d\ell_i|_{g_*} \le L$. Process all q coordinates with (15), using any smooth uniformly bounded graph fields with bounded derivatives and strict uniform tangent-band margins, not necessarily exact solutions. Require those fields to be compatible under the fixed bundle transition maps and invariant in the old circle directions. At step i, let e_i be its actual residual for the current physical metric. Put*

$$c_* = (1-a)^2 + a/2 < 1, \qquad K_q = \frac{c_*}{1-c_*}, \qquad \mathcal{E}(x) = \sum_{i=1}^{q} \{K_q e_i(x)^2 + 2(U_i e_i(x))_+\}. \tag{16}$$

Then every new coordinate obeys

$$\left|\ell_i^+ - \ell_i\right| < \pi w/2, \qquad \left|d\ell_i^+\right|_{g_h^+} \le L/4, \qquad g_h^+ \ge g_*,$$

and, with $C_0 = 128\sqrt{2}\, e^3$,

$$\mathrm{Scal}_{g^+} \ge \mathrm{Scal}_g - C_0(n+1)(2+\log q)L^2/w^2 - \mathcal{E}. \tag{17}$$

Uniform geometry is asserted at each fixed finite stage only.

Proof. The new coordinate is $\ell_i^+ = \varepsilon f_i$. Its height error is exactly $w\,|\theta_{f_i}| < \pi w/2$. Also $|df_i|_{g^+} = vt \le 1$: for $t < 1/2$ it is at most t, and for $t \ge 1/2$ one has $N = 1$. Later rank-one metric increases cannot enlarge these covector norms. This part does not use the graph equation.

Write $b_U = |U|_{g_*} \le r$. At the current step

$$A_+ \le (Lb_U - \varepsilon r t)_+, \qquad \lambda \ge B_0 r, \qquad b \le 2L/(wB_0) = a/2.$$

Thus $c \le c_*$. Notice that

$$1 - c_* = \tfrac{3}{2}a - a^2 > 0;$$

the residual coefficient grows at most on the order of $1/a$, rather than being independent of the coordinate count.

For $r < s$, $\lambda b_U \le \sqrt{2}B_0 s^{\beta+1} \le \sqrt{2}B_0/q$. For $r \ge s$, the adverse region requires $t \ge 1/4$ and $b_U > rt/4 \ge r/16$. Moreover

$$r^{-a} \le s^{-a} = \exp(3\log q/(2+\log q)) \le e^3.$$

Consequently $\lambda b_U \le 16\sqrt{2}e^3 B_0 b_U^2$ there. These are precisely the source's small- and large-slope estimates; they remain applicable because the adverse quantity has not been changed. Theorem 5.1 therefore gives

$$\mathrm{Scal}_{g^+} - \mathrm{Scal}_g \ge -\frac{C_0 L^2}{aw^2}(q^{-1} + b_U^2) - K_q e_i^2 - 2(U_i e_i)_+.$$

All inverse horizontal metrics act on the same original fibre cotangent space. The exact rank-one identity in (9) telescopes:

$$\sum_i U_i \otimes U_i = g_*^{-1} - (g_h^+)^{-1} \le g_*^{-1}, \qquad \sum_i |U_i|_{g_*}^2 \le n. \tag{18}$$

Sum the preceding scalar bounds and the q copies of $1/q$. No added circle direction enters this trace. Bounded derivatives and bounded gradients of the chosen fields ensure positive warp lower bounds and uniform metric equivalence at each finite stage. □

6.1 Summing Passes Without Hiding a Limit

For the source schedule $L_j = 4^{-j}L_0,\ w_j = 2^{-j}/10$, define $\mathcal{E}_j$ on the original base via the unchanged circle invariance. For every finite k,

$$\mathrm{Scal}_{\mathrm{fibre},k} \geq \mathrm{Scal}_{\mathrm{fibre},0} - \frac{400}{3}C_0(n+1)(2+\log q)L_0^2 - \sum_{j<k}\mathcal{E}_j. \tag{19}$$

The coordinate changes still total less than $\pi/10$. If one actually certifies $\sum_{j<k}\|\mathcal{E}_j\|_\infty \leq \eta$, the scalar lower bound loses at most this additional η. Individual bounds $\|e_{j,i}\|_\infty \leq \delta$ alone instead give as much as $kqK_q\delta^2$, before directional errors are counted. Neither a few successful coordinates nor an average residual implies the needed total supremum budget.

With initial curvature κ, for example, a source scalar cost at most $\kappa/8$, a residual cost at most $\kappa/8$, and a subsequent parameter-metric assembly cost at most $\kappa/4$ retain $\kappa/2$. The assembly cost is imported from the source's fixed-finite-stage mixed-derivative argument. It is not controlled by (7). The comparison scale R, and hence q and the parameter manifold, must first be fixed; only then is a sufficiently large finite pass count chosen. We construct no metric on an infinite-dimensional torus and take no uniform-in-k regularity limit.

7 Negative Controls and Finite Objects

7.1 Small Residual Does Not Control Scalar Curvature

On the flat two-torus with coordinates x, y, take $N = 1$, $\beta = 0$, $B_0 = \varepsilon = w = 1$, and

$$f_0(x,y) = \sin x, \qquad D_0 = -\partial_x \frac{f_0'}{\sqrt{1+(f_0')^2}}, \qquad \ell = f_0 + \arctan D_0.$$

This is fixed smooth bounded data, with a uniform tangent-band margin, and $F_\ell(f_0) = 0$. For positive integers j tending to infinity, put $f_j = f_0 + j^{-5/2}\sin(jx)$. Then $\|f_j - f_0\|_{C^2} = O(j^{-1/2})$ and smooth dependence of the flux divergence on two jets gives $\|F_\ell(f_j)\|_\infty \to 0$. The band margins remain positive. Nevertheless at $x = 0$, with $p_j = 1 + j^{-3/2}$,

$$f_j'' = 0, \quad D_j = 0, \quad \mathrm{II}_j = 0, \qquad \mathrm{Scal}_{g_j^+} = \frac{2p_j f_j'''}{(1+p_j^2)^2} = -\frac{2p_j(1+\sqrt{j})}{(1+p_j^2)^2} \longrightarrow -\infty. \tag{20}$$

Indeed the ordinary-graph identity is $\mathrm{Scal}_{g_j^+} = |\mathrm{II}_j|^2 + D_j^2 - 2U_jD_j$, and $U_jD_j = -p_jf_j'''/(1+p_j^2)^2$ at this point. Thus neither small C^0 residual nor C^2 closeness of the graph function controls scalar curvature of the stabilized metric. The directional derivative in (12) detects the missing third derivative. Taking $j = m^2$ makes the values in (20) rational, enabling exact finite checks.

7.2 The Actual Rank-One Metric Telescope

In two original directions start with inverse metric $P_0 = I_2$. For a covector a_i and a positive scalar $n_i = N(t_i)$, one update is

$$U_i = \frac{n_iP_{i-1}a_i}{\sqrt{1+n_i^2a_i^{\mathsf{T}}P_{i-1}a_i}}, \qquad P_i = P_{i-1} - U_iU_i^{\mathsf{T}}.$$

Take $a_1 = (1,0)^{\mathsf{T}},\ a_2 = (0,2)^{\mathsf{T}}$, and $n_1 = n_2 = 1$. Then

$$P_1 = \operatorname{diag}(1/2, 1), \quad P_2 = \operatorname{diag}(1/2, 1/5), \quad \sum_i |U_i|^2 = 1/2 + 4/5 = 13/10 \le 2.$$

The circle count is absent for a reason: the covectors vanish exactly on circle directions. If arbitrary graph fields depend on old circles, the updates are no longer confined to this two-dimensional block, and the claimed bound by $n = 2$ is unjustified. Approximate invariance is not exact invariance.

7.3 Finite Tests Versus a Function-Space Certificate

A finite weak observation bank can miss high-frequency residuals. For instance on a periodic coordinate circle, $\sin(jx)$ pairs to zero with every trigonometric polynomial of degree $< j$, while its derivative has amplitude j. Even exact vanishing of all those moments does not bound $\|Ue\|_\infty$. To apply Theorem 6.1 to a reconstructed field one needs certified bounds for the actual smooth residual and its directional derivative on every relevant point, together with transition, band, and geometry checks. Taylor remainders or interval enclosures may supply such bounds on a specified compact panel; a cover-wide certificate additionally requires uniform control over all panels.

8 Ram Formulations with Their Domains Retained

The freshly read original TR8 proof [9] uses a bounded Lipschitz domain, $V = H_0^1(\Omega)$, and a bounded Hermitian uniformly positive coefficient field. Its Lax–Milgram argument proves a weak inverse $V' \to V$, with the Dirichlet trace supplied by a lifting. The PDE–Green equivalence then follows by applying that actual domain-specific inverse. It does not produce a bounded global pointwise kernel or estimates on an exhaustion. The Neumann discussion adds compatibility and one constant-mode gauge per connected component. Here the positive height coefficient removes that constant mode in the bounded class; we do not add an arbitrary mean-zero gauge.

For our graph equation the weak residual on a compact domain is

$$\langle F_\ell(f), \varphi \rangle = \int_\Omega \langle U_f, d\varphi \rangle - \lambda(r_f) \tan\theta_f \, \varphi \, dV_g, \qquad \varphi \in C_c^\infty(\Omega). \tag{21}$$

If tests meet the boundary, the strong-residual pairing additionally contains $-\int_{\partial\Omega} \varphi \, U_f \cdot \nu \, dS$. Boundary values, or the boundary functional, must therefore be part of any Ram panel reconstruction. The graph flux is nonlinear in the gradient. A weak finite response matrix does not have the same inverse as the full operator (2); the drift due to $\lambda(r_f)$ is an explicit example of a term lost by freezing.

TR8's original quasilinear proof retains every coefficient variation in a Newton equation and requires a domain on which the products and derivatives are bounded. On a uniform positive-margin bounded-gradient neighborhood here, $X = C_b^{2,\alpha}$ and $Y = C_b^{0,\alpha}$ are appropriate for F_ℓ. A differentiable-residual scalar certificate instead needs additional regularity, for example a uniform $C^{3,\alpha}$ trial field with corresponding coefficient bounds. Those two uses must not be conflated.

The original TR1 stability theorem [10] assumes a genuine bounded isomorphism $B : Y \to X$, an admissible closed ball, and

$$\eta = \|BF(c)\|, \qquad q_B = \sup_{\text{ball}} \|I - BDF\| < 1, \qquad \eta + q_B r \le r.$$

It proves existence and uniqueness within that ball by contraction. All three conditions are required. A left inverse of a rectangular moment Jacobian does not establish them on $C_b^{2,\alpha}$. Its

transfer theorem likewise requires actual bounded coordinate and observation isomorphisms; finite truncations have no such property on the unrestricted function space.

Our height comparison takes a different route from that contraction theorem. It needs no small Lipschitz constant, but presupposes admissible fields and, for comparison with a root, existence of that root. Our scalar estimate uses the actual directional error rather than reinterpreting a coefficient norm as a curvature norm. This is the source-specific benefit of applying the Ram domain-and-residual discipline to this equation, not a new generic inverse theorem.

9 Source Architecture and Global Boundaries

Source module	Role and retained obligation
Introduction, reductions	Rational fundamental class, orientations, degree conventions, positive or Ricci-flat alternative.
Comparison	Ordered group-labelled complexes, finite equivariant carriers, compact pairing, relative realization and a nonzero local degree; $q = n + \dim A$, $\log q \leq JR$.
Graph calculus	Exact rank-one inverse, reciprocal circle volume, Jacobi sign and scalar identity (10).
Graph existence	Uniform cover charts, bounded solution class, tangent barriers, gradient bound, interior bootstrap, uniqueness and mixed-parameter gluing.
Shortening	Fibre row norms, fixed original rank n, finite passes, and only afterwards enlargement of the parameter metric.
Obstruction	Compact band and doubled degree map; scalar and full differential bounds on the entire inverse image of a target arc, independently of the finite circle count.
Final proof	First fix a sufficiently large R, hence q, A, Z; then choose a sufficiently large finite pass count k.

The comparison module does not assume contractibility of the universal cover. Its ordered auxiliary complex removes torsion stabilizers; fillings are made inside target carrier balls. Relative duality and rational realization produce the parameter manifold. Doubling may cancel a global count, so the retained local degree is over one half. The maximum-coordinate boundary gap is then preserved by shortening. These topological steps were read, but are not re-proved by our local PDE estimates or by the exact tests.

The terminal source input is Cecchini–Schick's Theorem 1.14(1) [3]: on a closed oriented manifold, a nonzero-degree map to $S^1 \times \mathbb{T}^{D-1}$, with scalar lower bound χ and full differential bound ϵ on the full preimage of a connected open arc J, satisfies

$$2\pi\epsilon \geq \sqrt{\chi}\,\mathrm{Length}(J).$$

The source retains that full-preimage condition when doubling its band. The published proof's high-dimensional minimal-slicing input remains imported. Neither a fibre-only differential estimate nor a single sampled part of the arc satisfies its hypotheses.

No source correction was required for the inspected PDE signs or shortening constants. The present changes are extensions and quantitative refinements, not repairs to an alleged missing global proof. Conversely, reading all source text does not independently certify every cited global input.

10 Current Public Status

The assigned September 23, 2026 manuscript is a public-release preprint claiming the general rational-inessentiality theorem [1]. Its cached text is studied here; captured-tree identity and online availability are not community validation. Chodosh's ICM 2026 survey, Section 8, still formulates the general aspherical statement as Conjecture 8.1 and records dimensions $2, 3, 4, 5$ in Theorem 8.3 [5]. We therefore do not present the source headline as an independently established conclusion of this report.

The publisher abstract for Chodosh–Li's 2024 paper records the four- and five-dimensional aspherical obstruction [4]. Ma–Wang–Xie–Yu–Zhu's August 28, 2026 version retains a spin universal-cover assumption for simplicial-volume vanishing [6]. Bi–Zhu's September 11 version describes singular dimension descent and enlargeable obstructions [7]. Di Cerbo–Dranishnikov–Jauhari's August 17 version proves its proposed extension in the spin case [8]. These are distinct scopes, not citations proving every manifold case asserted by the assigned manuscript. Our refresh checked their primary abstracts and version records, the stated torical theorem, and the specified survey section; it did not audit the full external proofs.

11 Advantages, Costs, and Further Work

The nonmonotone-profile extension isolates the real ellipticity condition and supplies admissible profiles excluded by monotonicity. Its cost is the need to bound the radial eigenvalue away from zero on the attained gradient interval. It does not authorize arbitrary profiles in the logarithmic pass, whose estimates also use the special small-slope cutoff and the q^{-3} parameter.

The arctangent bound is strictly smaller than its linear counterpart for a positive residual bound and has the correct small-error slope. Normalized residuals can exploit the actual local $\lambda(r)$ instead of its worst-case lower bound. It controls only height. Near a tangent-band endpoint, derivative estimates and computational conditioning can still deteriorate; no uniform parameter limit $\varepsilon \downarrow 0$ or $s \downarrow 0$ is supplied.

The two-square completion supplies a finite, explicit penalty for inexact graph solves and preserves the original-rank trace summation. Its cost is an actual cover-wide bound on e and Ue, exact circle invariance, and compatible transition data. The penalty is not uniform as $c \uparrow 1$. In particular, this report does not provide a verified numerical solver on an infinite cover or claim an implementation speedup.

A useful next step would certify a smooth local reconstruction together with its third-derivative remainder and a boundary-compatible global assembly. Another would optimize the pointwise c and the allocation of residual budgets across coordinates using the actual shortening directions. Any such allocation must bound the sum, not only each term separately. Extending to approximately invariant graph fields would require new estimates for the added circle directions; the rank-n telescope cannot simply be reused.

12 Read and Check Scope

The owned ledger records PRE hashes before numbered literal displays, their actual intervals, POST hashes, and unread complements. All 13 cached public text files were read: 11 TeX files, including all eight mathematical modules and both figures, plus the bibliography and README, totaling 3,006 lines. The public scientific reading is complete at that cached endpoint, but this is not a proof audit of every cited publication.

Fresh private-original reading covers TR8 lines 63–180 and 270–413, and all 295 lines of the TR1 stability module: 557 lines in total. TR8's other 804 lines and its separately included tables were not read for this report. No whole-corpus claim is made. The private files were used locally only and were never sent to a public search service.

The owned executable check uses exact rational arithmetic for 149 cases and 562 assertions. It checks the tangent-chord excess, radial and transverse signs, the complete two-square identity and its minimizers, pass margins, the worked budget, the periodic curvature jets, and actual two-dimensional inverse-metric updates. There are no floating-point diagnostics in this receipt and no parent or foreign cases in these counts. These tests supplement the displayed proofs; they certify no global PDE or topological theorem.

13 Conclusion

The auxiliary graph PDE admits a precise relaxation of profile monotonicity. Its tangent forcing also yields a nonlinear height comparison, while its scalar identity imposes a genuinely stronger directional residual requirement. Retaining that distinction gives an explicit inexact shortening theorem at fixed finite stages. The rational-essentiality comparison construction, parameter assembly, and terminal obstruction remain separate global obligations. This local analytic increment is the basis for C3; L3 describes applicability and is unchanged.

References

[1] OpenAI. *Positive scalar curvature forces rational inessentiality.* September 23, 2026. Cached source at commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Public source directory. All cached mathematical modules read; global assertions distinguished from local results here.

[2] D. Gilbarg and N. S. Trudinger. *Elliptic Partial Differential Equations of Second Order*, second edition, Classics in Mathematics, Springer, 2001. DOI. Classical Dirichlet, interior Hölder, and Schauder inputs used at the source's stated interfaces; book not freshly audited in full.

[3] S. Cecchini and T. Schick. *Enlargeable metrics on nonspin manifolds.* Proc. Amer. Math. Soc. 149 (2021), 2199–2211. Author version, Theorem 1.14(1); DOI. The statement was checked; its full proof and underlying minimal-slicing theory remain imported.

[4] O. Chodosh and C. Li. *Generalized soap bubbles and the topology of manifolds with positive scalar curvature.* Ann. of Math. 199 (2024), 707–740. Publisher abstract and record.

[5] O. Chodosh. *Minimal Surfaces and Comparison Geometry.* Proceedings of the ICM 2026, vol. 4, 82–102. Primary text, Section 8, Conjecture 8.1 and Theorem 8.3. Scoped statement reading.

[6] Q. Ma, J. Wang, Z. Xie, G. Yu, and B. Zhu. *Gromov's Simplicial Volume Vanishing Conjecture for Positive Scalar Curvature.* arXiv:2606.29135v4, August 28, 2026. Abstract and version record checked.

[7] Y. Bi and J. Zhu. *Positive Scalar Curvature Obstructions via Singular Dimension Descent.* arXiv:2606.20528v3, September 11, 2026. Abstract and version record checked.

[8] L. F. Di Cerbo, A. Dranishnikov, and E. Jauhari. *Symplectically aspherical Kähler manifolds, scalar curvature, and the fundamental group.* arXiv:2607.05170v3, August 17, 2026. Abstract and version record checked.

[9] M. Subbarao, original private Ram PDE corpus, `RamNonlin-IE-PDE-astra/TR8-lin-non-lin/TR8-lin-non-lin.tex`. Coercive inverse, boundary lifting, Green equivalence, semilinear and full quasi-linear derivative proofs; fresh literal ranges specified in Section 12. Confidential local source.

[10] M. Subbarao, original private Ram stability corpus, `RamNonlin-IE-PDE-astra/TR1-lin-nonlin/stability.tex`. Residual-ball proof, domain-preserving transfers, and boundary constraints. Entire module freshly read locally; no claim that other modules referenced there were read.